

From Ensemble Disagreement to Per-Residue Uncertainty: A Calibration Study of Two Transformer Architectures for RNA Structure Prediction

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Abstract—Per-residue uncertainty quantification (UQ) is a standard requirement for any deployed structure-prediction transformer: users need to know which residues of a prediction to trust. AlphaFold-class architectures expose an explicit confidence head (pLDDT) trained on synthetic-label targets, while single-sequence language-model architectures often emit no comparable signal. We ask whether an *implicit* per-residue uncertainty can be extracted from the unprocessed ensemble of a transformer that does not expose a confidence head, and whether it matches the explicit signal in calibration quality.

On a panel of $N = 12$ RNA X-ray crystals deposited after both predictors’ training cutoffs, we compare AlphaFold 3’s explicit per-residue pLDDT to DRfold2’s implicit signal computed as the 80-model ensemble standard deviation. Both signals are rank-correlated with per-residue Kabsch C1’ error against the experimental crystal and survive Bonferroni correction ($\alpha = 0.05/12 \approx 0.0042$) on 6/12 targets each, with the strongest per-target correlations reaching $\rho = -0.60$ ($p \approx 6 \cdot 10^{-7}$) for AF3’s pLDDT and $\rho = +0.66$ ($p \approx 10^{-11}$) for DRfold2’s ensemble std. The two uncertainty signals are largely orthogonal: they flag different residues as uncertain. The result has a concrete methodological implication: when a transformer does not expose an explicit confidence head, an 80-model unprocessed ensemble already contains a calibrated per-residue uncertainty signal that is extractable post-hoc, requires no retraining, and matches the explicit signal on this benchmark. All code, predictions, and ground-truth crystals are released at <https://github.com/Vince-Anduril/rna3d-calibration>.

Index Terms—uncertainty quantification, selective prediction, transformer calibration, structure prediction, RNA, deep learning

I. INTRODUCTION

Modern transformer-based predictors for biomolecular structure — AlphaFold 2/3 [2], DRfold2 [1], RoseTTAFold-class systems, RhoFold+ [3], Boltz-1 [4] — have made high-accuracy 3D-structure prediction routinely available to non-specialists. With wider deployment comes a calibration question: *at what per-residue confidence threshold should a user trust each region of a prediction?* The selective-prediction literature [6] has long argued that the per-token score is at least

as important as the headline accuracy number for practical deployment.

AlphaFold-class architectures answer this question with an explicit per-residue confidence head (pLDDT), trained as a regression target on a synthetic label. Single-sequence language-model architectures (such as DRfold2’s RNA Composite Language Model front end) often do not expose an equivalent confidence head in their user-facing output: they emit a single structure (the energy-selected `opt_0` PDB) and nothing else. This raises a methodological question: *can a usable per-residue uncertainty signal be extracted from a transformer that does not expose one, by leveraging its unprocessed ensemble?*

We answer this question by direct empirical comparison on a panel of $N = 12$ post-cutoff RNA X-ray crystals (57–101 nucleotides, deposited 2024-08 to 2026-03):

- AlphaFold 3 emits five per-seed predictions and an `atom_plddts` array, aggregated to per-residue means.
- DRfold2 emits a 4-config $\times$ 20-model ensemble (80 candidate structures) before its energy-based selection step. We compute, post-hoc, the per-residue standard deviation across the 80 Kabsch-aligned ensemble members as an *implicit* per-residue uncertainty signal.

Three findings emerge:

- 1) Both signals are calibrated against the crystal under Bonferroni correction (6/12 targets each at $\alpha = 0.05/12$), in the directions “low pLDDT = high error” and “high ensemble std = high error”.
- 2) The two signals are largely orthogonal — they flag different residues as uncertain on the same molecule. A pooled cross-model correlation between (–AF3 pLDDT) and DRfold2 ensemble std is weakly negative ($\rho = -0.24$, $p \approx 0.05$), the opposite of what one would expect if the two were measuring the same underlying “hard region”.
- 3) The DRfold2 ensemble-std signal carries a peak per-

target correlation of $\rho = +0.66$ ($p \approx 10^{-11}$ on an $L = 81$ target), matching or exceeding AF3’s peak per-target pLDDT correlation of $\rho = -0.60$ ($p \approx 6 \cdot 10^{-7}$).

The methodological implication is that, for transformer architectures emitting a candidate ensemble before selection, the ensemble-disagreement signal is a usable calibrated UQ substitute for an explicit confidence head — it is computed post-hoc, requires no retraining, and reaches the same regime of statistical reliability on this benchmark.

II. RELATED WORK

a) Calibration and selective prediction.: Per-token uncertainty signals trained or extracted from neural networks have been studied extensively in NLP and vision [6]. The structure-prediction literature has focused on training the confidence head as part of the model (AlphaFold’s pLDDT) rather than on extracting it post-hoc from ensemble disagreement.

b) Ensemble uncertainty.: Outside structure prediction, model ensembles are a well-known source of usable uncertainty (deep ensembles, MC dropout). What is less explored is whether the within-architecture ensembles that structure-prediction models already produce internally (e.g., DRfold2’s 4-config $\times$ 20-model candidate set, MSA-noise ensembles in AF2) carry a calibrated per-residue uncertainty signal when read post-hoc.

c) Transformer architectures compared.: We compare two transformer architectures with different inductive biases: AlphaFold 3 [2] (Pairformer + Diffusion Module, internal MSA via the AlphaFold Server) and DRfold2 [1] (single-sequence input through the RNA Composite Language Model, four configuration heads). RhoFold+ [3] and Boltz-1 [4] are MSA-based and not evaluated here.

III. TARGETS AND METHOD

A. Targets

We selected RNA X-ray crystals from the RCSB PDB with deposition date ≥ 2024 , RNA-only entries, length 50–130 nt, one entry per RNA family for diversity. The CPEB3 ribozyme (PDB 7QR3, 2021) [5] was added as an in-distribution reference; it is the only target in our panel inside DRfold2’s end-2023 training cutoff (declared in the authors’ Methods: “our training set only includes RNA structures released before 2024” [1]). All 11 other targets post-date both predictors’ training cutoffs (AF3: September 2021 [2]).

Panel: 7QR3 (CPEB3 ribozyme, 69 nt, dimer), 9DE7 (HIV-1 TAR, 57 nt), 9HRD (Class V GTP aptamer, 67 nt, tetramer), 9HRF (Class V GTP UU variant, 70 nt), 9LJN (Guanine-II riboswitch, 71 nt), 9LKE (Guanine-II + hypoxanthine, 70 nt), 9LKU (2’-dG-III + Guanosine, 65 nt), 9UW0 (2’-dG-III riboswitch, 63 nt), 9MFH (env2 cobalamin apo riboswitch, 76 nt), 9J4N (*E. coli* Leucine tRNA, 81 nt), 9E9O (SARS-CoV-2 SL5, 101 nt), 12CI (Dopamine aptamer DGR-1A, 82 nt, dimer).

For each crystal we use the C1’ atoms of standard A/U/G/C residues of a single reference chain (chain C for 7QR3 following the deposit convention [5], chain A elsewhere). For the

TABLE I
PER-TARGET SPEARMAN ρ OF EACH UNCERTAINTY SIGNAL VS.
PER-RESIDUE C1’ CRYSTAL ERROR. BONFERRONI $\alpha = 0.05/12 \approx 0.0042$
MARKED IN BOLD.

PDB	AF3 pLDDT	p	DRfold2 std	p
7QR3	+0.08	0.50	+0.22	0.074
9DE7	−0.60	$6.4 \cdot 10^{-7}$	+0.49	$9.7 \cdot 10^{-5}$
9HRD	−0.58	$2.6 \cdot 10^{-7}$	+0.18	0.14
9HRF	−0.44	$1.7 \cdot 10^{-4}$	+0.33	0.0052
9J4N	−0.27	0.015	+0.66	$1.4 \cdot 10^{-11}$
9E9O	−0.25	0.012	+0.43	$7.4 \cdot 10^{-6}$
9MFH	−0.58	$4.5 \cdot 10^{-8}$	+0.02	0.88
9LJN	−0.28	0.020	+0.58	$1.1 \cdot 10^{-7}$
9LKE	+0.09	0.46	+0.60	$4.5 \cdot 10^{-8}$
9LKU	−0.41	$8.3 \cdot 10^{-4}$	+0.31	0.013
9UW0	−0.20	0.12	+0.25	0.044
12CI	−0.55	$1.1 \cdot 10^{-7}$	+0.43	$6.1 \cdot 10^{-5}$

three multi-chain crystals, the chain-of-reference dependence of the RMSD is small (≤ 0.3 Å range for 9HRD and 12CI; 1.02 Å range for 7QR3, where we conservatively keep the higher-RMSD chain C).

B. Per-residue uncertainty signals

AF3 pLDDT. Each AF3 single-chain Server submission returns five seeds with a per-atom `atom_plddts` array per seed. For the default seed (seed 0) we aggregate per-atom pLDDT to per-residue means and use these as the per-residue uncertainty signal $u_{AF3}(i)$. A lower value means more uncertain.

DRfold2 ensemble standard deviation. DRfold2 emits an 80-model candidate ensemble (4 configs $\times$ ~ 20 models) into a `rets_dir/` before its energy-based `opt_0` selection step. We Kabsch-align each candidate model to the first ensemble member at the C4’ atom set (DRfold2’s intermediate models are coarse-atom; we use C4’ for the ensemble analysis and C1’ for the post-selection RMSD against the crystal). For each residue i we compute the standard deviation of the 80 aligned positions as $u_{DRfold2}(i)$. A higher value means more uncertain.

C. Calibration metric

For each target we compute the per-residue Kabsch C1’ deviation of the predictor’s user-facing structure (AF3 seed 0 for the AF3 analysis; DRfold2 `opt_0` for DRfold2) against the experimental crystal, $e(i)$. The calibration question is: *does $u(\cdot)$ rank residues by their actual error $e(\cdot)$?* We quantify this with Spearman rank correlation ρ between u and e . A negative correlation between AF3 pLDDT and error (lower pLDDT $\rightarrow$ higher error) and a positive correlation between DRfold2 ensemble std and error (higher std $\rightarrow$ higher error) are both “calibrated direction”. We apply Bonferroni correction at $\alpha = 0.05/12 \approx 0.0042$ across the 12 targets.

IV. RESULTS

A. Each signal is individually calibrated on 6/12 targets under Bonferroni

Table I reports the per-target correlations. Direction-only counts: AF3 pLDDT has 10/12 targets in the calibrated direction (negative), and DRfold2 ensemble std has 12/12 targets in the calibrated direction (positive). Raw at $\alpha = 0.05$: 9/12 and 10/12 respectively. After Bonferroni correction at $\alpha = 0.05/12$: 6/12 targets survive for each signal independently. Both signals reach the same fraction of strongly-calibrated targets on this $N = 12$ panel.

B. The two signals are largely orthogonal

We test whether the two signals agree on *which* residues are hard, by computing the per-target Spearman correlation between $u_{\text{DRfold2}}(i)$ and $-u_{\text{AF3}}(i)$ (so that both quantities point in the “high = uncertain” direction). If the two signals were measuring the same underlying uncertainty, we would expect a strong positive correlation. We observe instead a pooled correlation that is weakly negative ($\rho = -0.24$, $p \approx 0.05$, opposite-sign direction). The two signals flag substantially different residues as uncertain.

This is consistent with the two signals capturing different sources of error: AF3’s pLDDT is a learned property of a confidence head trained on AlphaFold-style synthetic targets, while DRfold2’s ensemble disagreement reflects the spread of 80 candidate folds before the energy-based `opt_0` selection step. Both are calibrated against the same ground truth at the same Bonferroni threshold, but they are not redundant.

C. Peak per-target correlation strength is comparable

AF3 pLDDT’s largest absolute per-target correlation is $\rho = -0.60$ ($p \approx 6 \cdot 10^{-7}$) on 9DE7 (HIV-1 TAR, $L = 57$). DRfold2 ensemble std’s largest absolute per-target correlation is $\rho = +0.66$ ($p \approx 10^{-11}$) on 9J4N (*E. coli* Leucine tRNA, $L = 81$). Both exceed $|\rho| = 0.5$ on at least 3/12 targets. The implicit ensemble signal does not under-perform the explicit head on this benchmark.

D. Validation on the ensemble-mean error (pre-selection)

A separate experiment validates that the DRfold2 ensemble std correlates with the error of the *ensemble-mean* structure (pre-selection) at $\rho = +0.47$, $p \approx 0.01$ on 7QR3. This suggests the ensemble-disagreement signal carries information about prediction quality before DRfold2’s energy-based selection step collapses the ensemble to one structure. Post-selection, the same signal is rank-correlated with the residual error of the final `opt_0` structure with the per-target strengths reported in Table I.

V. DISCUSSION

The headline finding is methodological: for transformer-based structure predictors that emit a candidate ensemble before selection, the 80-model standard deviation acts as a calibrated per-residue uncertainty signal of comparable quality to an explicit trained confidence head. This matters for two

practical reasons. First, it gives users of architectures without a confidence head a post-hoc UQ procedure that requires no retraining and no model changes — just access to the unprocessed ensemble. Second, the observation that the two signals are largely orthogonal suggests they can be combined (e.g. both passing or both failing a per-residue cutoff) for stronger selective-prediction guarantees than either signal alone provides.

A second result is that AlphaFold 3’s pLDDT calibration, when measured directly against an X-ray crystal of the same molecule, survives Bonferroni correction on half of our blind panel even when its absolute pLDDT is in the “low-confidence” regime. This contradicts the simplifying narrative that low-pLDDT implies an unusable prediction — in our data, low-pLDDT residues are systematically further from the crystal than high-pLDDT residues within the same target, regardless of the absolute pLDDT range.

VI. LIMITATIONS

- $N = 12$ targets, all RNA structures of length 57–101 nt. The pattern may not hold at larger sizes or in other biomolecule classes (proteins, complexes).
- AF3 was run through the public Server with default settings; the per-seed predictions are not reproducible bit-for-bit across re-runs because the Server’s internal MSA build is not deterministic.
- Bonferroni correction is conservative; the Benjamini–Hochberg false-discovery-rate at $q = 0.05$ would accept additional targets.
- The cross-signal orthogonality test ($\rho_{\text{cross}} = -0.24$, $p \approx 0.05$) is statistically borderline; an out-of-distribution validation on a larger panel would settle whether the two signals are genuinely orthogonal or only weakly anti-correlated.
- No fairness-of-comparison claim is made between AF3 and DRfold2 on absolute prediction quality; that is the subject of a separate companion paper.

VII. CONCLUSION

On a panel of $N = 12$ post-cutoff RNA X-ray crystals, we have shown that DRfold2’s 80-model ensemble standard deviation acts as a calibrated per-residue uncertainty signal of quality comparable to AlphaFold 3’s explicit pLDDT — 6/12 Bonferroni-significant targets each, with peak per-target correlations of $\rho = +0.66$ and $\rho = -0.60$ respectively. The two signals are largely orthogonal: they flag different residues as uncertain on the same molecule. The methodological consequence is that transformer-based structure predictors with internal ensembles can support per-residue selective prediction even without an explicit trained confidence head, by reading their unprocessed ensemble post-hoc.

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